

BAPIL

(BAPIL)

(AI) , -

- ,

: .

:

: (White Paper)

: - (Public Discussion Draft)

(Version 1.0)

BAPIL White Paper: Bangladesh AI Policy & Innovation Lab (Version 1.0)-

, ,

, -

(Disclaimer)

, , - ,

(BAPIL)- (Conceptual Framework)

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,

-

(Exploratory) ,

,

(Suggested Citation)

, ()

BAPIL : (.)

BAPIL Research Initiative.

Zenodo. DOI: (To be assigned).

(Acronyms)

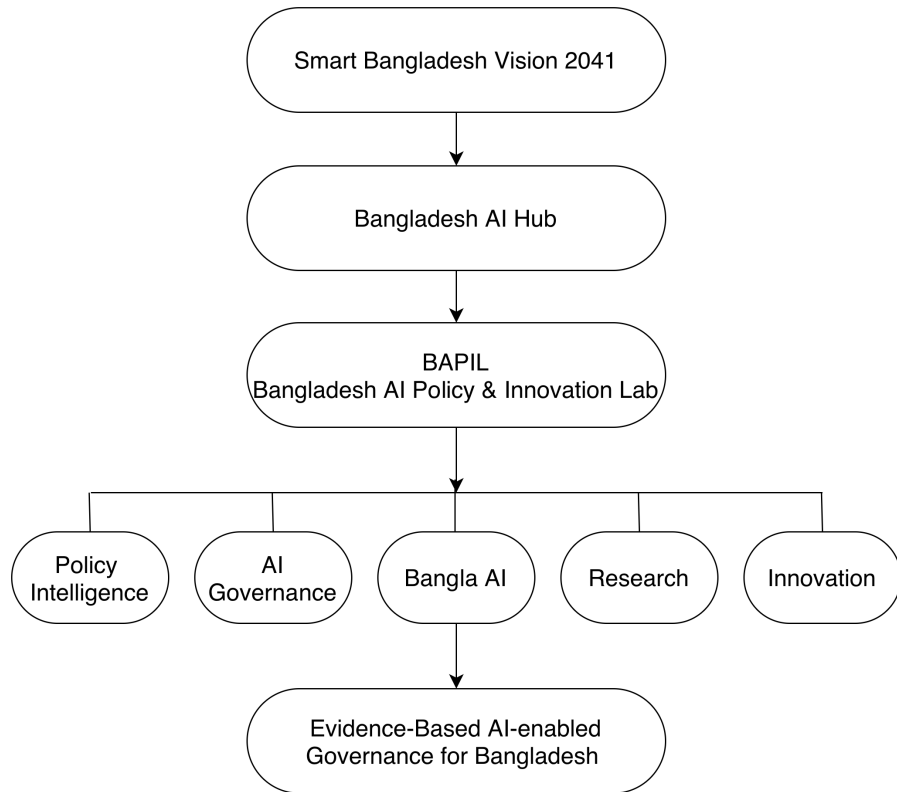
AI
NLP
RAG - -
BPFM
BAPIL
OECD
NIST ()
UNESCO ,
WEF
UNDP
ITU
BCC
BBS
ICT
GovTech
LLM
XAI

(Glossary)

, , , , , ;
 ;
 ,
 , (BAPIL)- ;
 , - ,
 , ,
 :
 (Artificial Intelligence - AI) , , , ,
 AI- ,
 [1-17, 21, 26]
 -
 , -
 [28-33, 82, 83]
Bangladesh AI Policy & Innovation Lab (BAPIL)-
 (Conceptual Research Framework) ,
 , AI ,
 AI , ,
 ,
 [1, 3, 5, 7, 17, 21, 24]
 BAPIL- (Multidisciplinary) , , ,
 - - (Policy Intelligence), , AI ,
 , [17-27, 71-77]
 :
 • AI (AI Policy & Research)
 • (Data Analytics & Public Sentiment Analysis)
 • (Digital Security & Misinformation Detection)
 • AI (AI Governance & Ethics)
 • AI (AI Education & Talent Development)
 • (Innovation & Startup Collaboration)
 BAPIL- ,
 , AI ,

AI
74, 75]

[1, 5, 6, 7, 15, 21, 24,



AI Bangladesh AI Policy & Innovation Lab
(BAPIL)-

:

-
-
-
-
-
-
-

(AI), AI

,

. **AI** - **(Policy Intelligence)**

AI- , (Evidence Synthesis) - (Decision Support)

[10, 17, 18, 20, 21, 26]

, AI- **BAPIL**

. **AI**

AI- AI

[1, 4-9, 14, 15, 51, 53, 68, 69]

AI :

-
-
- AI
- (Algorithmic Bias)
-
-

AI

. **AI**

[16, 27, 35, 36, 54-56, 63, 70, 73, 77]

:

- AI
- - (High-Performance Computing)
-
-
- -

,

. **AI**
 , , (NLP), (Speech Technologies)
 [37-50, 80, 81]
 :
 •
 • (NLP)
 • (Speech)
 • (Machine Translation)
 • - AI
 •
 AI ,
 . **(Open Data)**
 (High-quality Data) AI
 (Open Data) , (Interoperability) , (Metadata
 Standards) AI -
 [18, 20, 31, 33, 78, 79]
 :
 • (Data Quality)
 • (Open Government Data)
 • (Interoperability)
 • (Metadata Standards)
 • - (Data Sharing)
 •
 , AI ,
 . , ,
 [3, 17, 19, 24, 25, 27, 35, 36]
 :
 •
 • -
 •
 •
 • (Multi-stakeholder)
 • (Technology Governance)

. **AI**
Smart Bangladesh Vision 2041 **Bangladesh AI Hub-**
 AI [29, 30, 35, 36]
 :
 • - (Policy Research)
 • (Governance Innovation)
 • (Knowledge Management)
 • (Strategic Foresight)
 • - (Evidence-Based Decision Support)
BAPIL- - ,
 . **AI** **(Foresight)**
 (Technology Foresight),
 (Strategic Scenario Analysis), (Emerging Technology Assessment)
 (Future Workforce Planning)- [22, 23, 60, 63,
 73, 77]
 :
 • (Technology Foresight)
 • (Strategic Scenario Analysis)
 • (Emerging Technology Assessment)
 • (Future Workforce Planning)
 • AI- (AI Impact Evaluation)
 • - (Innovation Policy Research)
 , (National
 Resilience)
 ,
 , , ,
 AI ,
 ,
Bangladesh AI Policy & Innovation Lab (BAPIL)-
 , - (Policy Intelligence) ,

,
:
AI- - (Policy Intelligence)

, (Artificial Intelligence - AI),
(Advanced Analytics)
(Evidence-Based) [10-21, 26]
Digital Bangladesh Smart Bangladesh-
(Connectivity)
[28-33, 82, 83]

(Data-Driven)

, , , , ,
[18, 20, 21, 31, 33]
; (Actionable Insights)-
AI- - (Policy Intelligence)
[10, 17, 18, 20, 21, 24]

, (Stakeholders)
[10, 12, 17, 18]
AI- , (Knowledge Discovery)
[10, 17, 21, 38, 41, 42]

, (Online Consultation),
[17, 21, 25, 82, 83]
AI- (Sentiment Analysis) (Public Opinion Monitoring)
,

, [7, 24, 45, 80, 81]

(Misinformation), (Disinfor-
 mation), (Information Integrity) [14,
 34, 71, 72, 74, 75]

AI- (Information Verification) (Digital
 Resilience) [14, 71, 72, 74, 75]

AI

AI- (Transparency), (Accountability),
 (Fairness), (Privacy Protection) (Human Oversight)
 (Governance Mechanisms),
 AI AI [1, 4-9, 15, 51, 53, 68,
 69]

(Best Practices) AI
 [1, 5, 6, 15,
 51, 54, 57, 60, 68]

AI- - (Policy Intelligence) :

- [10, 17, 21]
- [17, 20, 21, 57]
- [10, 20, 68, 69]
- [18, 31, 33, 78, 79]
- [21, 25, 82, 83]
- [11, 17, 20, 57, 58]
- AI [35, 36, 70, 73, 77]
- , [16, 26, 35, 36, 60]

BAPIL-
Bangladesh AI Policy & Innovation Lab (BAPIL)-

, , **BAPIL** (Interdisciplinary)
 [3, 17, 19, 24,
 25, 35, 36]

[29, 30, 35, 36] AI- [10, 17, 21]

Bangladesh AI Policy & Innovation Lab
(BAPIL)
 [1, 3, 5, 17, 21, 24, 29, 30]

:

(Artificial Intelligence - AI), (Data-Driven) AI
 [10-21, 26]
 , AI , , - (Policy Intelligence)-
 [51-70]

• : **(Smart Nation) GovTech**
Smart Nation GovTech Singapore-

[51-53]

:

- (Centralized)
-
-
- AI (Responsible AI Governance)
-

AI Verify Model AI Governance Framework-
 , AI [51, 53]

-
-
-
- AI

Smart Bangladesh AI
[30, 51-53]

. : AI AI Hub

,
National AI Strategy- AI-
, [54-56]

:

- AI
- AI-
- ,
- AI (Data)
- AI

AI Hub ,
[54-56]

- AI
- AI
- - -
-

AI Hub AI [35, 36, 54-56]

. :

Governance Digital-First
[57-59]

:

- (Digital Identity)
- (Digital Signature)
- X-Road (Interoperability)
-
- (Citizen-Centric)

,
[57-59]

- (Interoperable)
-

•
•
- [32, 33, 57-59]

. : **AI**

AI ,
[60-62]

:

- **Government Digital Service (GDS)**
- AI
- **AI Safety** **AI Assurance**
- AI
- AI

[60-62] ,

- (Risk-Based) AI
- AI (Standards) (Best Practices)
-
- AI (AI Oversight)-

OECD, UNESCO Council of Europe AI
[1, 5, 15, 60-62]

. : **AI** **AI**

(AI) ,
AI
[63, 64]

“Elements of AI”, (Online Course)
AI ,

[63, 64] , ,

:

- AI
- (Digital Literacy)-
-
- AI
- ,

- AI AI
- AI
-
- (Interdisciplinary) AI

AI
[63, 64, 73, 76, 77]

. (UAE): AI
(United Arab Emirates - UAE)

for Artificial Intelligence, Minister of State
AI [65-67]
, (Citizen Experience) UAE
[65-67]

UAE- :

- AI
-
- AI-
- (Smart City)
-

UAE- AI
[65-67]

-
- AI-
-
-

(Technology Policy)
[29, 30, 65-67]

. : AI
AI AI AI
AI-
[68-70]

Directive on Automated Decision-Making,
AI- AI [68-70]

- - AI (AI Oversight)
 - AI
 -
 - (Risk-Based) AI
-
- AI (Standards)
 -
 - AI
 -
- AI [68-70]
- , [51-70]:
- [52, 54, 56, 60, 65, 66]
 - (Interoperable) [17, 20, 21, 57-59],
 - AI (Risk Management) [51, 53, 60-62, 68, 69],
 - [19, 27, 54, 63, 70]
 - AI AI [63, 64, 73, 76, 77]
- AI- [1, 3, 5, 10, 17, 21, 51-70]
- [51-70]
- AI -
- (Policy Intelligence), [1, 3, 5, 17, 21, 29, 30, 35, 36]
- AI AI [29-36, 80]

: **AI**

(Artificial Intelligence - AI),
, , , , ,
AI-
[24, 26, 35, 36, 80]
AI
AI
[28-33, 35, 36, 82, 83]

Digital Bangladesh Smart Bangladesh-

, - ,
[28-30, 32, 82, 83]

:

-
-
-
- (Cloud-Based) (Data-Driven)
-

AI [17, 20, 21, 28-30, 82, 83]

Vision 2021- **Digital Bangladesh**
- (ICT) [28, 29, 82]

Smart Bangladesh Vision 2041

:

- (Smart Government)
- (Smart Economy)
- (Smart Society)
- (Smart Citizens)

AI-
[29, 30]

[32, 82, 83]

(National Portal)

(Bangladesh National Portal)

(Unified Digital Gateway)

, [83]

AI- :

- (Structured Government Information Repositories)
- (Metadata)
-
- (Knowledge Management)

- **(e-Service)**

[82, 83]

:

-
- (Immigration)
- (Tax)
-
-

(Operational Data) , ,
[18, 20, 21]

- **(e-Nothi)**

- **(e-Nothi)** [32]

:

- (File)
- (Electronic Workflow)
-
- (Document Tracking)
-

- (Institutional Knowledge Repository) , - (Policy Intelligence)
(Knowledge Management) [32]

(Data)

[18, 20]

(Bangladesh Bureau of Statistics - BBS)

(BBS)

:

-
-
-
-
-
-
- (Sustainable Development Indicators)

,

[31]

(Open Data)

Bangladesh Open Data Portal

[33, 78, 79]

- (Policy Intelligence) AI

[18, 33, 78, 79]

(Digital Identity)

(Digital Identity)

[82, 83]

:

- (National Identity - NID)
-
- (Electronic Verification)

[17, 20, 21, 82]

:

- (Mobile Financial Services)
-
-
- (Financial Inclusion)

AI-

[29, 30, 82]

GovTech **GovTech (Government Tech-**
nology)- [17, 20, 21]
 :
 • (Digital Service Innovation)
 • (Public Sector Modernization)
 • (Government Innovation Laboratories)
 • (Citizen Engagement)
 GovTech AI- - (AI-enabled Policy Intelligence)
 [17, 20, 21, 32]

:
 • (Data Fragmentation)
 • (Interoperability)
 • (Standards)
 • - (Machine-Readable)
 • (Advanced Analytics) AI-
 • (Data Governance)
 AI- [18, 20, 31,
 33, 71, 72, 78, 79]

BAPIL-
 - (Policy Intelligence)
 [28-33]

BAPIL :
 •
 • - (e-Nothi)
 •
 • (Open Government Data)
 •
 • GovTech
 AI, (Analytics), (Knowledge Management) -
 (Policy Intelligence) [17,
 18, 20, 21, 32, 33]

, AI-
 , AI-

[30, 35, 36, 80]

:

- (Smart Government Services)
- (Data-Driven)
- (Digital Governance)
- (Digital Resilience)
- AI-
-

AI [35, 36] AI

BAPIL

Smart Bangladesh Vision 2041,
Bangladesh AI Hub- AI- [29, 30, 35, 36]
Bangladesh AI Policy & Innovation Lab (BAPIL)-
(Duplication) , - (Policy Intelligence), AI (AI
Governance) - (Evidence-Based Decision Support)
[1, 5, 7, 17, 24]
Bangladesh AI Hub- , , AI
(Policy) (Governance)- BAPIL AI- (Analytical),
[35, 36]
AI, (Knowledge-Based) :
•
•
• AI
• [1, 5, 7, 17, 18, 24]
, (Complementary) :

Smart Bangladesh Vision	
2041	
Bangladesh AI Hub	AI ,
BAPIL	- (Policy Intelligence) AI (AI Governance)

, BAPIL-
(Analytical Insights), AI (Governance Frameworks) (Knowledge
Resources) AI

, **BAPIL**,
AI

(Data Science), (Robotics) AI, (Machine Learning),
[35, 36, 80]

:

- AI
- (Natural Language Processing - NLP)
- (Computer Vision)
- AI
- (AgriTech)
-

, [35, 36, 73, 77, 80]

(Technology)
AI- (AI-enabled) :

- (FinTech)
- - (E-commerce)
- (HealthTech)
- (EdTech)
- (AgriTech)
- (Customer Service Automation)
- (Data Analytics) (Business Intelligence)

, (Incubators), (Accelerators) (Venture
Financing) AI- [26, 35, 36]

AI

:

- (Natural Language Processing - NLP)
- (Speech Recognition)
- (Machine Translation)
- (Information Retrieval)
-
- (Educational Technologies)

AI , (Language Resources)
(Digital Inclusion),
[46-49, 80, 81]

AI- , AI

:

- AI
- AI
- (Large-Scale Computing)
- (Data Quality) (Data Governance)
- (Cybersecurity)
- (Ethical) (Regulatory)
-

, , [7, 14, 35, 36, 71-80]

AI (Strengths) :

-
-
-
-
-
- AI

, , AI [26, 29, 30, 35, 36, 73, 77, 80]

BAPIL-

AI AI- - (Policy Intelligence)
AI- [1, 5, 7, 17, 24, 35, 36]
(AI Governance)

Bangladesh AI Policy & Innovation Lab (BAPIL)-
(Research-Oriented) , AI , - ,
 ,
BAPIL AI
(Evidence-Based Approaches) [3, 17, 19, 24, 25, 35, 36]

AI [29-36, 71-80]

AI

Bangladesh AI Policy & Innovation Lab (BAPIL)-

AI [1, 5, 17, 24, 29, 30, 35, 36]

: BAPIL-

(Artificial Intelligence - AI), (Data Analytics),
(Responsible AI Adoption), (Evidence-Based Policymaking),
(Public Sector Innovation) (Digital Governance)
[1-7, 10, 17-26]

Bangladesh AI Policy & Innovation Lab (BAPIL)- (Multi-disciplinary)
(Data-Driven Technologies), AI (Policy Intelligence),
(Digital Resilience) [3, 17, 21, 24-27]

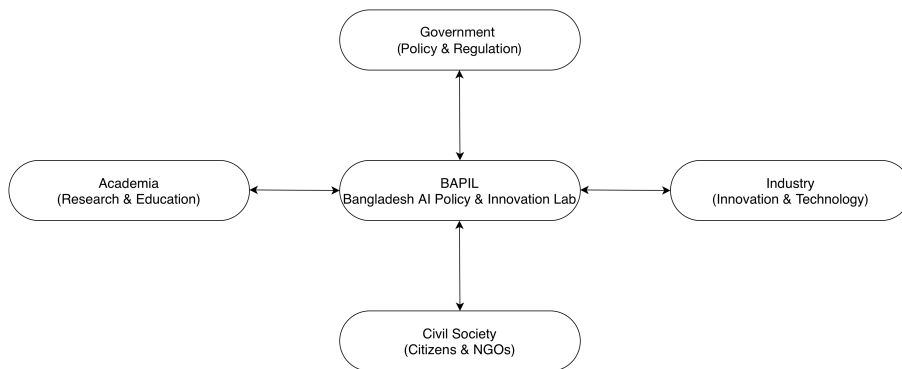
(Vision Statement)

(Transparent), (Ethical), (Trustworthy),
(Evidence-Based)
[1, 4, 5, 7, 15]

(Strategic Purpose)

BAPIL (Knowledge Generation),
(Interdisciplinary Collaboration) AI
[3, 17, 19, 24, 25, 27, 29, 30]

BAPIL- AI,
(Analytical Insights),
[10, 17, 18, 20, 21]



BAPIL- (Conceptual) , , , [19, 24, 25, 27]

BAPIL- AI , (Complementary)
Bangladesh AI Hub- , AI , AI
BAPIL , - , AI ,
 [35, 36]
 AI - [29, 30, 35, 36]

(Core Principles)

BAPIL-

(Human-Centered AI)

AI , (Human Oversight) [1, 5, 7, 15, 51, 68] ,

(Transparency)

, (Explainable) [5, 7, 50, 68, 69]

(Responsibility)

AI , [1, 4, 5, 15, 24, 74, 75]

(Inclusiveness)

AI- , (Equitable Digital Transformation) [3, 22, 23, 24, 76]

- (Evidence-Based Decision Support)

[10, 17, 18, 20, 21]

(Collaboration)

AI [19, 24, 25, 27, 35, 36]

(Strategic Objectives)

BAPIL-

- AI- (Policy Intelligence) [10, 17, 20, 21]
- AI [1, 4-9, 15, 51, 53, 68, 69]
- [17, 18, 20, 21]
- AI [46-49, 80, 81]
- (Misinformation) [14, 34, 71, 72, 74, 75]
- AI [35, 36, 63, 64, 73, 76, 77]
- [19, 24, 25, 27]
- [16, 26, 35, 36, 70]

(Potential Areas of Impact)

BAPIL AI

(Governance and Public Policy)

- [10, 17, 20, 21]
- [17, 20, 21]
- - [18, 20, 21]

(Education)

- AI (AI Literacy) [64, 76]
- [35, 36, 63, 73]
- (Educational Technology) [24, 76]

(Healthcare)

- (Health Data Analytics) [18, 24]
- (Public Health Intelligence) [17, 24]
- AI [16, 77]

(Agriculture)

- [18, 23]

- [22, 23]
- (Smart Farming) [24, 77]

(Digital Governance)

- [17, 20, 21, 57]
- [21, 25]
- [17, 18, 57, 58]

(Long-Term Aspirations)

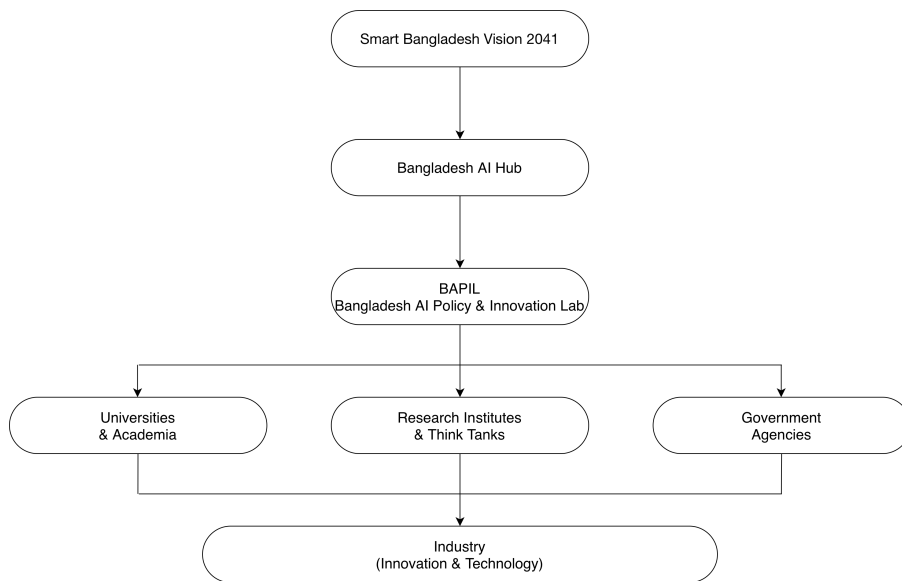
BAPIL AI -
 , AI [23, 29, 30, 35, 36, 73, 77]
 BAPIL (Policymaking Authority)
 ,
 [1, 5, 7, 15, 24]

(Vision for International Collaboration)

AI , BAPIL
 ,
 [3, 19, 24, 27, 51-70]
 , , AI
 [19, 24, 27, 51-70]

(Institutional Alignment with National Initiatives)

Bangladesh AI Policy & Innovation Lab (BAPIL)-
 - [29, 30, 35, 36]
 BAPIL , - , AI
 ,
 [17, 20, 21, 29, 30, 35, 36]
 :



Smart Bangladesh Vision 2041, Bangladesh AI Hub Bangladesh AI Policy & Innovation Lab (BAPIL)-

[29, 30, 35, 36]

Smart Bangladesh Vision 2041 [29, 30]
Bangladesh AI Hub AI [35, 36]
BAPIL - , AI [1, 7, 10, 17, 21]

BAPIL AI [19, 24, 25, 27, 35, 36]

BAPIL- [1, 5, 15, 23, 24]
 BAPIL AI- [29, 30, 35, 36]

: BAPIL- (Architecture)

Bangladesh AI Policy & Innovation Lab (BAPIL)-
 (Research-Oriented) - (Policy Intelligence)

(Data Analytics), (Artificial Intelligence - AI),
 (Knowledge Management) - (Decision Support)
 (Architecture), , (Digital Governance)
 AI- (Responsible AI Experimentation)
 [1, 4, 10, 17, 21]
 (Automated Policymaking)
 (Analytical Tools), (Knowledge Resources) AI-
 (AI-Assisted Insights) , ,
 [5, 7, 15, 17]

(Architectural Principles)

BAPIL- :

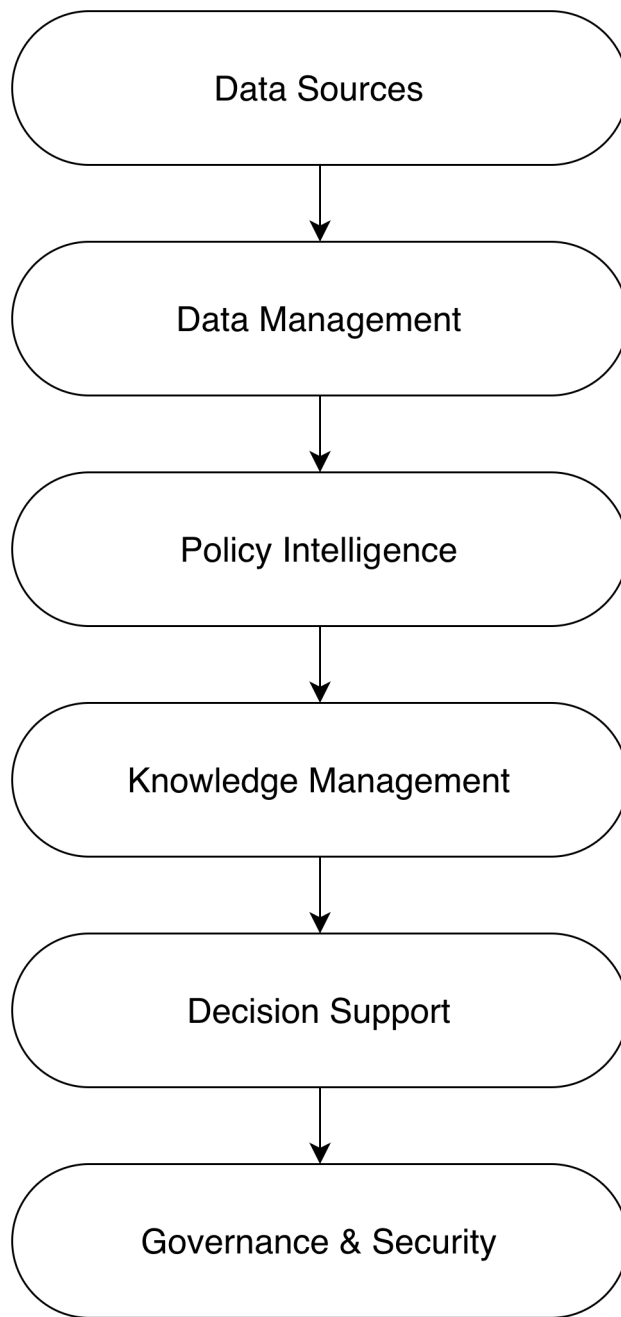
- - (Human-Centered Decision Support) [1, 4, 5, 15]
- AI (Responsible and Ethical AI) [2, 5, 6, 7]
- (Transparency and Explainability) [4, 5, 7, 50]
- (Data Security and Privacy Protection) [6, 14, 34, 71]
- (Interoperability) (Scalability) [11, 17, 20, 57, 59]
- (Research-Driven Innovation) [16, 24, 27, 35]
- (Modular) (Extensible) [7, 8, 10, 17]

- (High-Level Architecture)

BAPIL Architecture (Layers) :

1. (Data Sources Layer)
2. (Data Management Layer)
3. - (Policy Intelligence Layer)
4. (Knowledge Management Layer)
5. - (Decision Support Layer)
6. (Governance and Security Layer)

(Integrated Environment) , - ,
 AI- - [7, 10, 17, 18]



Bangladesh AI Policy & Innovation Lab (BAPIL)- (High-Level Conceptual Architecture), (End-to-End Flow)

(Evidence-Based Policymaking)

(Data Sources Layer)

Data Sources Layer

:

(Government Data)

- [31]
- [29, 31]
- [29, 31]
- [29, 30]
- (Public Service Datasets) [33, 83]

(Legislative and Policy Documents)

- [34]
- (Parliamentary Proceedings)
- [29, 30]
- [29, 30]

(Public Information Sources)

- (News Media)
- (Public Consultations)
- [24, 79, 80]
- (Open Government Data) [33, 78]

(Citizen Feedback Sources)

- (Surveys)
- [79]
- (Social Media Trends) [81]
- [17, 21]

[18, 21, 22]

(Data Management Layer)

Data Management Layer , , (Integration) -
(Preprocessing)-

:

- (Data Ingestion) [18, 20]

- (Data Cleaning) [18]
 - (Data Validation) [18, 31]
 - (Metadata Management) [18, 33]
 - (Data Cataloging) [17, 18]
 - (Secure Storage) [14, 34, 71]
- (Centralized Data Repository) ,
- (Data Governance), (Privacy) (Security)-
[14, 17, 18, 34]

- (Policy Intelligence Layer)

Policy Intelligence Layer BAPIL (Analytical)

:

(Policy Analytics)

- (Trend Analysis) [10, 17, 20]
- (Comparative Analysis) [17, 21]
- (Policy Impact Assessment) [10, 17]
- (Scenario Exploration) [60, 61]

(Natural Language Processing - NLP)

- (Document Summarization) [38, 39, 42]
- (Information Extraction) [38, 46]
- (Topic Modeling) [46, 81]
- (Sentiment Analysis) [81]

(Data Analytics)

- (Statistical Analysis) [18, 31]
- (Predictive Modeling) [39, 42]
- (Visualization) [17, 20]
- (Pattern Identification) [18, 77]

(Knowledge Discovery)

- (Identification of Emerging Issues) [2, 77]
- (Evidence Synthesis) [18, 24]
- (Cross-Sector Analysis) [17, 21]

; - [5, 7, 15]

(Knowledge Management Layer)

(Knowledge Resources)

Knowledge Management Layer- :

- (Policy Document Repositories) [32, 83]
- (Research Databases) [24, 79, 80]
- (Best Practices) [11, 17, 25]
- (Knowledge Graphs) [38]
- (Institutional Memory Systems) [32, 57]

, , (Institutional Learning)
(Knowledge Continuity) [17, 21, 32]

- **(Decision Support Layer)**

Decision Support Layer - (User-Facing) ,
(Analytical Outputs) ,

:

- **(Policy Intelligence Dashboard)**

- (Key Indicators) [20, 31]
- (Trend Monitoring) [17, 21]
- (Data Visualization) [20]
- (Research Insights) [24, 77]

(Interactive Analytics Tools)

- (Scenario Exploration) [60, 61]
- (Comparative Analysis) [17, 20]
- (Custom Reporting) [21]

(Knowledge Search Systems)

- (Intelligent Document Retrieval) [38]
- (Semantic Search) [38, 46, 49]
- AI- (AI-Assisted Information Discovery) [39, 42, 43]

(Usable) , (Accessible), (Transparent) [10, 17, 21]

(Governance and Security Layer)

AI (Oversight)

:

AI (AI Governance Framework)

- (Transparency Principles) [1, 4, 5]
- (Accountability Mechanisms) [5, 6, 15, 68]
- (Human Oversight) [5, 6, 68]

(Data Governance Framework)

- (Data Quality Standards) [18, 79]
- (Privacy Protection) [6, 14, 34]
- (Data Access Policies) [18, 33]

(Security Controls)

- (Cybersecurity Monitoring) [14, 71, 72]
- (Access Management) [14, 34]
- (Risk Assessment) [2, 7, 71]

(Cybersecurity), AI- (Ethical), (Legal),
(Institutional) [5, 6, 7, 14, 15]

(Future Research Components)

BAPIL- :

- (Bangla Policy Foundation Models - BPFM) [41, 46, 48, 49]
- Retrieval-Augmented Generation (RAG) [38]
- (Knowledge Graph) [38]
- - (Multi-Agent) [41, 42]
- AI (Explainable AI - XAI) [5, 7, 50]
- (Advanced Public Sentiment Analysis) [46, 81]
- - (Policy-Oriented Bangla Large Language Models - LLMs) [46, 48, 49]
- AI (AI Readiness) AI (Governance Benchmarking Systems) [26, 80]

[2, 5, 7, 50]

BAPIL Architecture (Conceptual),
(Data), (AI), (Knowledge Management) - (Policy
Intelligence) (Analytical Tools)
(Responsible Governance Principles),
(Public Sector Innovation) [1, 5, 7, 17, 21]

AI , (Digital Government),
(International Best Practices)-
, **Smart Bangladesh Vision 2041**, AI (AI Readiness)
AI [17, 20, 21, 29, 30, 35, 36, 80]

: - (Policy Intelligence Dashboard)

Bangladesh AI Policy & Innovation Lab (BAPIL)-
- (Policy Intelligence Dashboard)
(Integrated)

, (Policy Insights)
[10,17,21,24,25]

- (Decision Support)
(Knowledge Discovery) , (Data Visualization),
(Analytics) AI- (AI-Assisted Insights)
[1,4,10,17]

(Purpose of the Dashboard)

,
[17,18,20,21]

Policy Intelligence Dashboard- :

- -
-
-
-
-
- [17,20,21,25]

(Key Functional Components)

(Integrated)

(National Indicators Module)

[21,23,29,31]

:

- (Economic Growth)
- (Inflation)
- (Employment Statistics)

- - (Education Indicators)
 - - (Healthcare Indicators)
 - (Agricultural Production)
 - (Digital Transformation Metrics)
 - (Sustainable Development Indicators)
- (Interactive Visualizations)
[21,23,31]

(Policy Monitoring Module)

[11,12,20,32]

:

- (Policy Status Tracking)
 - (Performance Monitoring)
 - (Milestone Visualization)
 - (Program Evaluation Support)
 - (Strategic Planning Dashboards)
- (Visible), [11,20,32]

(Public Sentiment and Citizen Feedback Module)

[18,21,24]

:

- (Public Opinion Monitoring)
- (Survey Analysis)
- (Sentiment Analysis)
- (Citizen Feedback Aggregation)
- (Emerging Issue Identification)

AI- (Ethical Safeguards) [5,7,80,81],

(Policy Research Module)

[17,19,25,27] - (Research Resources)

:

- (Research Repositories)
- (Policy Briefs)
- (Comparative Studies)
- (International Best Practices)
- (Evidence Synthesis Tools)

[17,25,27] (Knowledge Sharing) (Policy Learning)

AI- (AI-Assisted Analytical Capabilities)

AI [37-45]

:

(Document Summarization)

AI , (Concise Summaries)

[39,42,43]

(Information Extraction)

(Large Document Collections)

[38,42,46,48]

:

- (Key Policy Themes)
- (Stakeholder Positions)
- (Strategic Objectives)
- (Regulatory Changes)

(Trend Analysis)

(Trends) (Patterns)

[18,21,77]

(Knowledge Discovery)

AI- , (Relationships)

[38,41,50]

, ; - [1,4,5]

(Visualization and User Experience)

- , ,

[17,20,21]

:

- (Interactive Charts)
- (Geographic Maps)
- (Policy Timelines)
- (Comparative Dashboards)
- (Network Visualizations)

- (Trend Monitoring Panels)
- (User Interface) - (User-Centered Design)
- [17,21]

(Knowledge Search and Retrieval)

[18,25]

:

- (Semantic Search Systems)
- AI- (AI-Assisted Information Retrieval)
- (Policy Document Repositories)
- (Knowledge Graph)
- (Cross-Document Linking)

[38,46,48,49]

(Stakeholder Perspectives)

[3,17,21,25]

(Policymakers)

- (Policy Insights)
-
-

(Researchers)

- (Datasets)
-
-

(Government Agencies)

- (Performance Indicators)
-

(Civil Society and Public Stakeholders)

-
-

(Transparency and Responsible Use)

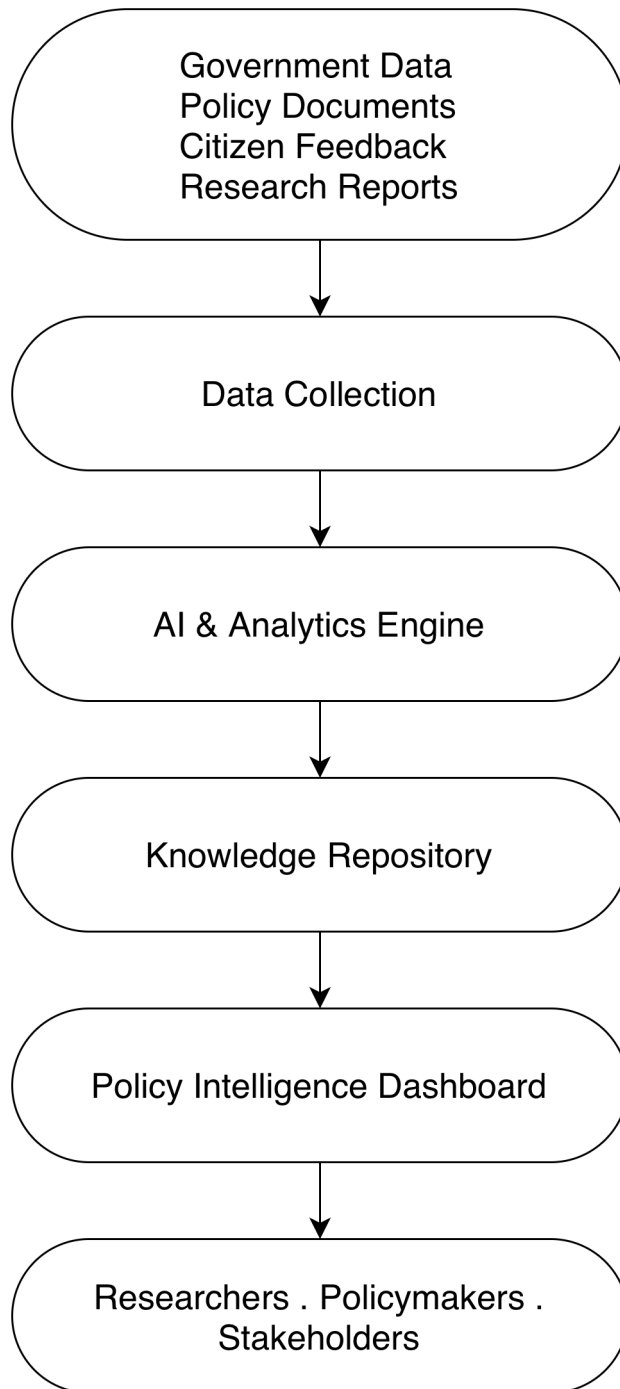
AI (Responsible AI) (Good Governance)

[1,4,5,6,7,15]

- (Transparency of Methodologies)
 - (Human Oversight)
 - (Data Privacy Protection)
 - (Security Controls)
 - AI (Ethical Use of AI Technologies)
 - (Explainability of Analytical Outputs)
- (Public Trust), (Accountability) AI
[1,4,5,6,7,14,15,74,75]

(Conceptual Dashboard Architecture)

- (Policy Intelligence Dashboard)
- :



Bangladesh AI Policy & Innovation Lab (BAPIL) - (Policy Intelligence) (Government Data), (Policy Documents), (Citizen Feedback) (Research Reports) (Data Collection), AI- (AI-Powered Analytics), (Knowledge Management) - ,

(Actionable Insights) (Accessible Visualizations)- [10,17,18,21,33,83]

(Future Opportunities)

[38,41,42,49,50]

:

- (Real-Time) (Policy Monitoring)
- AI- (AI-Powered Policy Assistants)
- (Bangla-Language Policy Search Systems)
- **Retrieval-Augmented Generation (RAG)-**
- (Policy Simulation Environments)
- (Explainable AI Analytics)
- (Cross-Sector Knowledge Graphs)

, , AI (AI Governance) [2,5,7,50,74,75]

(Conclusion)

Policy Intelligence Dashboard BAPIL (Data Analytics), AI- (AI-Assisted Insights), (Visualization Tools) (Knowledge Management)- (Evidence-Based Decision-Making) [17,21,24]

(Research and Decision-Support Platform)

[1,4,5,10]

Bangla Policy Foundation Model (BPFM)-

AI - , [37,46,48,49,80]

: (Bangla Policy Foundation Model - BPFM)

(Artificial Intelligence - AI)-
(Foundation Models)-
 [37,39,41,42,43,44]
 (Information Retrieval), (Doc-
 ument Analysis), (Knowledge Management), - (Decision
 Support) [10,17,21,24]
 - (High-Resource)
 , -
 [41,45,46,48,49]

(Bangla Policy Foundation Model - BPFM)

AI ,
 - (Policy Intelligence), - (Governance Research)
 (Public Sector Innovation)- [24,30,35,80]
 - **(Long-Term Research Direc-**
tion) ; (Implementation Plan)

(Foundation Model) ?

AI , (Large and Diverse Datasets)
 (Pre-trained) - **(Fine-**
Tuning) (Adaptation) [41]
 [39,42,43,44]:

- (Question Answering)
- (Text Summarization)
- (Information Retrieval)
- (Document Classification)
- (Language Translation)
- (Knowledge Extraction)
- (Conversational Assistance)

, - [10,17,21]

?

AI [46,48,49,80], -

(BPFM) :

- (Policy Documents) [29,30]
- (Government Reports) [31,32]

- - (Budget and Planning Documents) [29,31]
 - (Legislative Texts) [34]
 - (Citizen Feedback and Public Consultations) [21,24,81]
 - (Public Sector Knowledge Management) [17,21,82]
- , [18,21],

(Strategic Objectives)

Bangla Policy Foundation Model (BPFM)- :

- (Policy Intelligence)

- (Policy Document Analysis) [10,17]
- (Regulatory Monitoring) [6,15]
- (Legislative Analysis) [34]
- (Evidence Synthesis) [21,24]

(Knowledge Management)

- (Government Knowledge Repositories) [17,21]
- (Research Databases) [25]
- (Institutional Memory Systems) [32]
- (Public Information Access) [33,83]

(Citizen Engagement)

- (Analysis of Public Feedback) [21,24]
- (Sentiment Analysis) [81]
- (Public Consultation Support) [22]
- (Citizen Information Services) [17,21]

(Bangla Language Technology)

- (Bangla Natural Language Processing - NLP) [46,48,49]
- (Document Understanding) [41,42]
- (Question-Answering Systems) [39,42]
- (Semantic Search) [38]

(Potential Data Sources)

,

(Government Publications)

- (Policy Documents) [29,30]

- (Strategic Plans) [29,30]
- (Budget Reports) [31]
- (Annual Reports) [31]
- (Regulatory Documents) [34]

(Parliamentary and Legislative Resources)

- (Parliamentary Proceedings)
- - (Legislative Records) [34]
- (Committee Reports)

(Research and Academic Materials)

- (Research Publications) [25]
- (Policy Studies) [17,21]
- (Development Reports) [23,24]
- (International Best Practices) [11,17,21]

(Public Information Resources)

- (News Archives)
 - - (Public Consultation Documents) [21,22]
 - (Open Government Datasets) [33,78,79]
- , (Governance)

[1,4,5,6,15,34]

(Conceptual Architecture)

Bangla Policy Foundation Model (BPFM)-
 grated Components)

(Inte-

(Data Collection Layer)

(Trusted) (Governed) - ,
 [18,33,78,79]

(Language Processing Layer)

:

- (Tokenization) [37]
- (Language Understanding) [39,42]
- (Named Entity Recognition - NER) [46]
- (Information Extraction) [41]

(Knowledge Layer)

:

- (Knowledge Repositories) [17,21]
- (Knowledge Graphs)
- (Policy Document Indexing)
- (Semantic Relationships)

(Foundation Model Layer)

:

- (Language Understanding) [41,42,43,44]
- (Context-Aware Retrieval) [38]
- (Question Answering) [39,42]
- (Document Summarization) [41]

(Application Layer)

:

- - (Policy Intelligence Dashboards) [17,21]
- (Research Tools)
- (Search Systems) [38]
- (Analytical Interfaces)

Retrieval-Augmented Generation (RAG)

Retrieval-Augmented Generation (RAG)

[38]

Knowledge Repositories), **RAG** (External [38])

:

- (Improved Factual Grounding) [38]
- (Better Transparency) [50]
- (Access to Updated Information) [38]
- (Hallucination) [2,38]
- [17,21]

, (Traceability), RAG-
[2,7,15]

(Potential Use Cases)

Bangla Policy Foundation Model (BPFM)

(Policy Document Search)

(Natural Language Queries)

[38]

(Policy Summarization)

,

[41,42]

(Budget Analysis)

AI-

,

[31]

(Legislative Analysis)

,

[34]

(Knowledge Discovery)

,

,

(Relationships)

[17,21]

AI (Responsible AI Considerations)

-

AI (Responsible AI)-

[1,4,5,6,7,15,51,53]

:

- (Transparency) [1,4,50]
- (Human Oversight) [5,15,68]
- (Data Governance) [18,21]
- (Privacy Protection) [5,34]
- (Security Controls) [14,71,72]
- (Fairness and Bias Mitigation) [45,50]
- (Accountability Mechanisms) [1,15,68,69]

;

-

[1,4,5,7]

(Research Challenges)

,

:

- [46,48,49,81]
- (Data Standardization and Quality) [18,78,79]
- [35,36,77]
- [41,50,77]
- [1,5,6,15]

• [17,24,30]

[24,27,35,36,80]

(Long-Term Vision)

Bangla Policy Foundation Model (BPFM) AI

[24,30,35,36,76,80]

BAPIL- (Knowledge Discovery), -
 (Policy Intelligence)
 AI [1,4,5,24,30]

(Conclusion)

Bangla Policy Foundation Model (BPFM) - ,
 AI - (Policy Intelligence) (Public
 Sector Innovation)- [24,30,35,80]

(Bangla Language Processing), (Knowledge Manage-
 ment), (Information Retrieval) **AI (Responsible AI)**
 [38,46,48,49], (Evidence-Based Governance)
 (Digital Transformation)- [17,21,30]

AI Governance Framework , BAPIL
 AI (Institutional, Ethical and Regulatory)
 [1,4,5,6,15]

: AI (AI Governance Framework)

(Artificial Intelligence AI) (Governance),
 (Public Administration), (Research) - (Decision Support
 Systems)- AI (AI Governance)
 [1,4,5,6,7,10,15] AI , (Evidence-Based Policy-
 making) (Transparency), (Accountability),
 (Privacy), (Bias), (Security) (Public Trust)-
 [2,5,6,7,14,15]

BAPIL **AI Governance Framework-**
 (Oversight Mechanisms) , AI- (Development),
 (Deployment) (Evaluation)- [1,4,5,7]

AI (Public Interest)-
 (Legal, Ethical and Institutional) [5,6,15,17,21]

AI (AI Governance) ?

AI Governance (Policies), (Processes), (Standards)
(Institutional Mechanisms)- , - ,
[1,4,5,6,7]

AI :

- (Responsible Innovation) [1,4,62]
- [5,6,68]
- [5,15]
- [2,7]
- [1,5,15]
- [4,7,68]
- , AI [5,6,15]

AI ;
[1,4,5]

(Guiding Principles)

BAPIL AI Governance Framework AI
, - (Policy Intelligence) (Public Sector
Innovation) [1,4,5,7,15]

(Transparency)

AI- - [1,4,5,50,68]

:

-
- (Explainable Outputs)
-
-

(Accountability)

AI- [1,5,6,15,68]

:

-
-
- (Auditability)
-

(Fairness and Inclusion)

AI (Bias)
[5,15,45,53]

:
 •
 •
 •
 •
 (Privacy Protection)
 , [5,6,15,34]
 :
 • (Data Minimization)
 •
 • (Controlled Access)
 • -
 (Human Oversight)
 AI ; - [1,5,6,51,68]
 :
 •
 •
 •
 • AI- (Override Mechanisms)
 (Safety and Reliability)
 AI , [2,6,7,14]
 :
 •
 •
 •
 •
 (Governance Structure)
 AI (Complementary Layers)-
 (Strategic Governance Layer)
 :
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 •

- AI AI [1,9,15,54,60,70]

(Technical Governance Layer)

:

- AI
-
-
-

NIST AI Risk Management Framework (AI RMF), OECD-Responsible AI- [7,8,10,14]

(Ethical Governance Layer)

:

-
- Responsible AI
-
-

UNESCO Council of Europe- [5,15]

(Operational Governance Layer)

:

-
-
- (Incident)
-

AI [17,20,21,61]

AI (AI Lifecycle Governance)

AI (Lifecycle)- [7,14]

(Data Collection)

-
- Data Governance-
-

(Model Development)

-
- -
-

(Deployment)

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(Monitoring)

-
-
-

(Retirement and Updating)

-
-
- (Responsible Decommissioning)

(Risk-Based Governance Approach)

AI Risk-Based Governance
[6,7,8,68,69]

- (Low-Risk Systems)

- :
- -
 - (Visualization)

- (Medium-Risk Systems)

- :
- -
 -

- (High-Risk Systems)

- :
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- AI
- - AI
- , [6,7,15,68,69]

(Data Governance Framework)

AI Data Governance [18,21,78,79]

:

-
- Data Stewardship
- Metadata Management
-
-
-

[18,21,33,78,79]

(Explainability and Transparency)

- (Policy Intelligence)- AI [1,5,7,50,68]

:

-
-
- (Assumptions)
-
- (Confidence Level)

Explainable AI (XAI) [5,7,50]

(Ethics Review and Impact Assessment)

BAPIL- AI (Structured Ethical Review Process) [5,15,53,69,74]

:

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[5,15]

(International Alignment)

BAPIL AI Governance Framework

:

- OECD AI Principles
- UNESCO Recommendation on the Ethics of AI
- NIST AI Risk Management Framework
- European Union AI Act
- Singapore Model AI Governance Framework

AI

[1,5,6,7,51]

AI (Governance and Public Trust)

AI (Public Trust) [1,5,15,74,75]

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AI (Trustworthy) ,
[1,5,15]

BAPIL- (Role of BAPIL)

BAPIL- AI Governance (Foundational Pillar)
, (Responsible Innovation) - (Policy Intelligence)
[24,25,35,36,80]

BAPIL-

- AI -
- (Best Practices)
-
- AI
- (Interdisciplinary)

BAPIL- AI (Regulation) ; ,
AI - [24,25,27]

(Conclusion)

(AI) , AI
AI Governance Framework

[1,5,6,7,15]
 (Transparency), (Accountability), (Fairness), (Privacy), (Security) (Human Oversight)-
BAPIL AI Governance Framework BAPIL AI ,
 [1,4,5,6,7,15]
 (National Security) (Cybersecurity)-
 (Information Integrity) (Digital Resilience),
 [14,34,71,72]

:

, (Artificial Intelligence - AI), -
 [2, 14, 17, 21, 71, 72]
 (Digital Resilience),
 [14, 17, 71, 72]
 [29, 30, 82, 83] , (Information Integrity),
 [34, 71, 72, 80]
 BAPIL , ,
 [30, 35, 36]

[14, 71, 72] , ,
 , [17, 20, 21, 71]
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, ; , [14, 17, 21, 71]

[2, 14, 71, 72]

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-
- (Data Breach)
- (Misinformation) (Disinformation)
-
- (Technology Supply Chain)
- AI-

 , [21, 30, 71, 72, 80]

(AI)
 [2, 7, 14, 75]

AI-
 AI
 :

-
- (Anomaly)
-
- (Fraud)
-
-

 [7, 14, 71, 72, 77]

AI-
 , AI , :

-
- (Phishing)
- (Synthetic Media) (Deepfake)
- AI-
- AI (Adversarial)

 AI [2, 6, 7, 41, 45, 74, 75]

(Digital Resilience)
 , , [17, 20, 21, 71]

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[20, 21, 71, 72]

(Information Integrity)

, (Misinformation) (Disinformation)
[2, 45, 74, 75]

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AI- [2, 5, 45, 74, 75]

[17, 20, 21, 71]

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17, 20, 21, 71] [14,

[71, 72, 73]

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80] [35, 36, 73, 76, 77,

(Responsible Security Governance)

[5, 6, 7, 14, 15] ,

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(Proportionality)

[5, 6, 15, 68]

71, 72] ; [19, 27,

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-

(Best Practices)

71, 72, 75] [19, 27,

BAPIL-

BAPIL , - (Policy Intelligence)

[30, 35, 36, 80]

:

- AI
-

• (Information Integrity)
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 BAPIL- ; ,
 [30, 35, 36]

:
 • AI- [7, 14, 71]
 • [14, 34, 68]
 • [45, 74, 75]
 • [20, 21, 71]
 • AI (Secure AI Systems) [2, 6, 7]
 • (Privacy-Preserving Technologies) [5, 15]
 • [5, 6, 14, 15]
 AI, [2, 7, 75, 80]

[29, 30, 35, 36, 80]
 , [14, 34, 71, 72]

BAPIL- , - (Evidence-Based)
 [30, 35, 36, 80]

(AI Education and Talent Development)
 , AI ,
 ,

: ,

(Artificial Intelligence - AI) , [1, 5, 6,
 21] , ,
 [23, 26, 73, 77]
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 , ,
 - [23, 26, 63, 73]

BAPIL ,

[1, 5, 7, 14, 76]

[4, 9, 23, 26, 77]

[28, 29, 31, 80]

[30, 32, 35, 36]

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(AI Readiness) [26, 29, 35, 36, 79, 80]

(Lifelong Learning)

[23, 73, 76]

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[5, 23, 73, 76] UNESCO-
[76]

[23, 26, 63, 77]

• ;
• ;
• ;
• (Interdisciplinary) ;
• ;
•

[29, 30, 35, 63, 77]

[64, 73]

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• - - ;
• ;
•
[23, 64, 71, 73] , -

[41, 50, 77]

• ;
• (High-Performance Computing) ;
• ;
• ;
• ;
•

[2, 29, 35, 41, 77]
, (Low-Resource Lan-
guage Technologies) [46, 47,
48, 49, 80]

[10, 11, 17, 21]

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- ;
- Policy Intelligence;
- ;
- ;
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[10, 11, 17, 21, 32, 82]

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- ;
- (Information Systems)

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[1,

5, 6, 15, 45]

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[5, 7, 15, 45]

[2, 23, 26, 27]

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[2, 23, 27, 63, 77]

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[51-56, 58, 60-70]

BAPIL-

BAPIL

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[1, 17, 21]

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[1, 5, 7, 15]

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[23, 35, 36, 73, 77]

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[17, 23, 26, 27]

BAPIL-

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[29, 35, 36, 46-49, 77, 80]

[29, 30] , , , ,

[23, 26, 29, 30, 77]

[23, 26, 73, 76] [1, 5, 17, 21] ,

BAPIL- ,
 , , [1, 5, 23, 29, 77]

, ,

:

(Innovation) (Entrepreneurship) ,
 , [10-13, 17-20, 26, 73] (AI)-

(Artificial Intelligence - AI) [1, 4, 9, 17, 24, 26, 54, 70] - , [28-30, 35, 36]

BAPIL

(Innovation Laboratory) , ,
 - [10, 17, 19, 26, 50] ,

- :
 - (Economic Diversification) [17, 18, 26];
 - (Technology Commercialization) [41, 50];
 - [73];
 - [17, 21, 28];
 - (Research Translation) [41, 50];
 - (Export-Oriented Innovation) [26]
- [41, 50]

[28-30, 32, 82, 83] , -
[28, 29, 32, 82]

- (FinTech);
 - - ;
 - ;
 - (HealthTech);
 - (EdTech);
 - (AgriTech);
 -
- ,
[28, 30, 82]

[1, 4, 17, 24, 37-44]

- :
- [10, 17, 21];
- [17, 21];
- [18, 21];
- Policy Intelligence [1, 4, 9]

:

- - ;
- (Health Data Analytics);
- ;
-

- :
- (Smart Farming) ;
- ;
- ;
-

- :
- (Personalized Learning Systems) [39, 42, 76];
- - [39, 42];
- (Educational Analytics);
- [46-49]

- :
 - (Fraud Detection);
 - (Risk Assessment);
 - (Financial Inclusion) ;
 - (Intelligent Customer Support Systems) [39, 42]
- ,
- [4, 17, 24, 26]

19, 26, 35, 36] (Stakeholders) [17,

(Entrepreneurs)

, [41, 50]

, , [17, 29, 30, 35, 36]

(Investors)

, [26]

(Industry Partners)

,

, , [26, 35, 36]

, [17, 19, 26]

- [17, 26, 35, 80]

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- [26];
- [35, 73, 77];
- [41, 50];
- [18, 33, 78, 79];
- [1, 5, 6];
- [35, 36];
- [26]

, , [17, 19, 26, 35, 36]

(Research Commercialization)

, [41, 50]

:

- - (University Spin-offs);
- (Technology Transfer) ;
- - [35, 36];
- (Applied Research) ;
-

[41, 50]

[1, 5, 17, 21, 24]

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- [10, 17, 21];
- [24, 76];
- ;
- [23, 24];
- (Digital Inclusion) [3, 18, 22]

[1, 5, 24]

BAPIL-

BAPIL ,

:

- [17, 26];
- (Innovation Policy) [9, 17];
- (Technology Adoption) [4, 10]

- (Best Practices) ;
- ;
-

- ;
- (Interdisciplinary) ;
- [17, 35, 36]

(Policy Intelligence)

- [1, 4, 9];
- [26];
-

BAPIL ,

[51-70]

:

- [63, 64, 70];
- [54, 58, 70];
- [26];
- [6, 51, 62, 68];
- [27, 51, 53];
- [52, 57]

, , , , (UAE)
[51-70]

(Long-Term Vision)

- [29, 30, 35, 36]
:

- [18, 29];
- [73, 77];
- [26];
- [35, 36];
- [17, 21, 28]

[18, 29, 35, 73, 77]

,

[17, 26, 35, 36]

BAPIL , (Policy Intelligence)

[1, 3, 23, 24]

(Budget Estimation) , BAPIL-

, ,

: -
 (BAPIL)-
 BAPIL-
 (Phased Implementation) ,
 [1, 3, 9, 17, 22, 29, 30]
 (Policy Intelligence)
 [1, 4, 10, 17, 18, 26]
 - (Human Capital) [2, 24, 26, 73, 77]
 BAPIL- :
 • (Machine Learning Researchers) [37-44];
 • (Data Scientists) [24, 26, 77];
 • (AI Engineers) [2, 7, 42];
 • (Natural Language Processing - NLP) [46-49, 81];
 • (Knowledge Graph) [37-41]
 • (Policy Analysts) [1, 10, 17, 21];
 • [3, 11, 25];
 • [12, 21];
 • (Development Researchers) [18, 23]
 • [14, 34, 71, 72];
 • (Digital Risk Analysts) [7, 14];
 • (Information Security Specialists) [14, 71]
 • [41, 50];
 • [18, 33];
 • (Knowledge Management) [17, 18]

- ;
 - ;
 -
- BAPIL- (Interdisciplinary) , -
[1, 5, 15, 17, 25]
- Policy Intelligence [17, 18, 20, 21, 30] , ,
- :
- (Cloud Computing Platforms) [16, 35, 36];
 - (Research Servers) [35, 77];
 - (AI Development Environments) [37-44];
 - (Data Processing Systems) [17, 18]
- (Secure Data Repositories) [18, 33, 78, 79];
 - (Knowledge Management Systems) [17, 21];
 - (Metadata Management Tools) [18];
 - (Data Integration Platforms) [17, 20]
- Policy Intelligence [10, 13, 17];
 - [17, 18];
 - [37-44];
 - (Information Retrieval Tools) [38, 46-49]
- (Access Control Mechanisms) [14, 34];
 - [5, 14];
 - [14, 71];
 - (Backup and Recovery) [14]
- (Scalability), (Security) (Sustainability)-
[7, 14, 17, 71]

Policy Intelligence , [18, 21, 33, 78, 79]

:

- [29, 30, 83];
- [31];
- (Policy Documents) [1, 29, 30];
- (Legislative Records) [34];
- [41, 50];
- (Open Government Data) [33, 78];
- (Public Consultation Reports) [79];
- [1, 4, 5, 6, 15]

[1, 5, 7, 15, 18]

BAPIL- [17, 19, 27, 35, 36]

:

- [17, 21, 30];
- [1, 9];
- [10, 17, 32];
- [35, 36]

:

- [41, 50];
- [16, 27];
- [35, 36];
- (Knowledge Exchange) [24, 25]

:

- ;
- [16, 35];
- [35, 36];
- [41]

- (Policy Dialogue) [22];
- (Public Engagement) [5, 15];
- (Independent Analysis) [79];
- (Knowledge Dissemination) [25]

BAPIL- ,
[17, 19, 27, 35]

, [24, 35, 73, 76, 77]

- (AI Literacy) [24, 64, 76];
 - Policy Intelligence [1, 10];
 - (Research Workshops) [35, 36];
 - (Fellowship Programs) [16, 27];
 - (Student Engagement) [64, 76];
 - (Professional Development) [73]
- ; (Strategic Investment)
[3, 24, 73, 76]

BAPIL

- [30, 32];
- [35, 36];
- [29];
- [35]

- ;
- (International Research Grants) [16, 27];
- (Collaborative Research Initiatives) [19, 27]

- (Multilateral Organizations) [17-25];
 - (Development Agencies) [3, 24];
 - (International Cooperation Programs) [16, 36]
- :
- (Research Sponsorship);
 - (Technology Partnerships);
 - [16, 35]
- (Diversified Funding Strategy) (Resilience)
- (Sustainability) [17, 19, 24, 27]
- BAPIL (Phased Implementation)
- [17, 29, 30]
-
- (Literature Review) [1-16];
 - (Comparative Studies) [51-70];
 - (White Paper) ;
 - (Stakeholder Engagement) [22, 25]
-
- (Architecture) [37-44];
 - Policy Intelligence Dashboard- [10, 17];
 - (Research Demonstrations) [41, 50]
-
- [35, 36, 76];
 - [27, 35];
 - (Knowledge Sharing) [25]
-
- [16, 19, 27];
 - [26, 35];
 - [41, 50]

[17, 24, 29] , BAPIL-

-

BAPIL- :

- (Research Excellence) [41, 50];
- (Institutional Credibility) [1, 5];
- [17, 19, 27];
- [35, 73, 76];
- [17, 24];
- (Demonstrated Public Value) [10, 17, 23]

(Sustainability)-

[3, 17, 24]

BAPIL- , , [1, 3, 7, 17, 18, 35]

(Collaborative) BAPIL

, Policy Intelligence, (Responsible) (Evidence-Based)

[1, 4, 5, 7, 10, 15]

(Five-Year Roadmap) ,

BAPIL- Policy Intelligence Innovation Framework-

: : (-)

Bangladesh AI Policy & Innovation Lab (BAPIL)- , BAPIL , [1-5, 9, 17, 21]

- (Conceptual) ,

BAPIL- Policy Intelligence Innovation Framework- [1, 9, 10, 17, 21]

(-)

BAPIL- , AI Governance, Policy Intelligence,

[1, 4, 9, 17, 22, 29, 30]

[28-30, 35, 36]

BAPIL-

AI Governance Framework,
Policy Intelligence

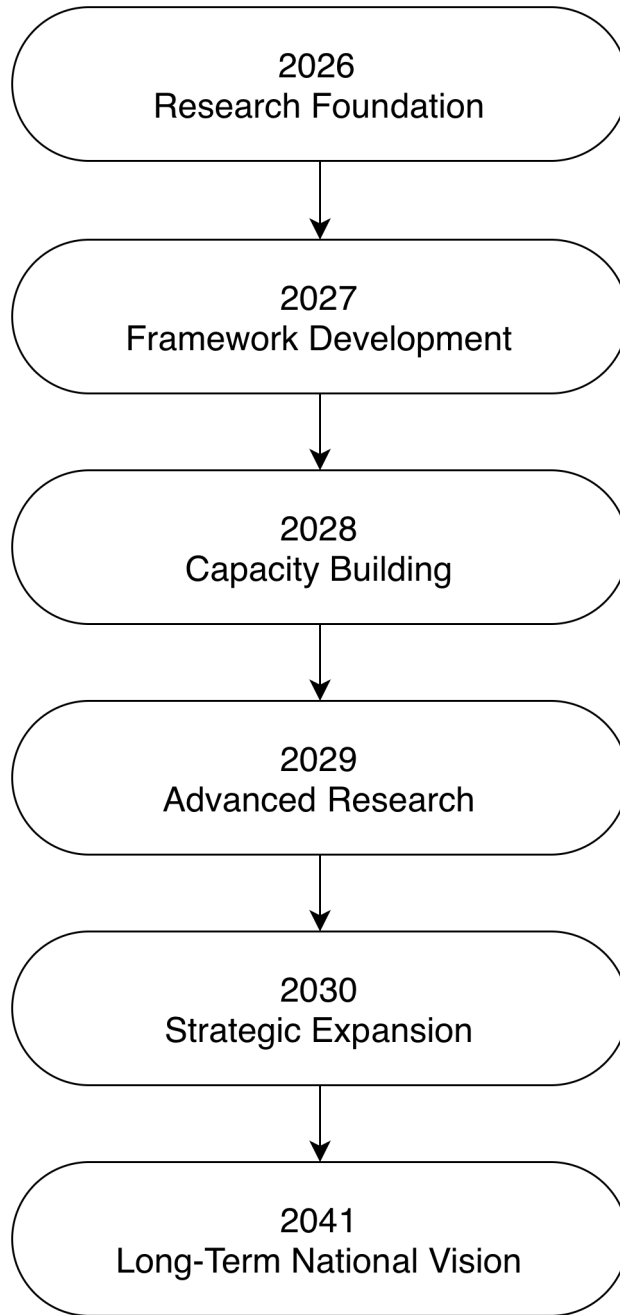
,

AI , AI
Policy Intelligence

,

,

AI-



Bangladesh AI Policy & Innovation Lab (BAPIL)-

, , , ,

AI [1, 4, 5, 9, 17, 21, 24, 76] , , ,
 BAPIL , [1, 5, 7, 15, 22, 74, 75]
 (Flexible) (Adaptive) , ,
 9, 22, 29, 30] [2, 8, AI
 [9, 10, 15]

[1, 4, 5, 9, 17, 29, 30]:

- AI Governance Policy Intelligence-
-
- AI
- AI
- , ,
-

: ()

- BAPIL-
-
-

- BAPIL White Paper
- AI Governance Policy Intelligence [1-10, 15]
- , , , , (UAE) [51-70]
- (Technical Architecture)
- [24, 27, 35, 36]
- (Stakeholder Consultations) [5, 22, 76]

-
-
- BAPIL-

: ()

-
- (Analytical)

- BAPIL Policy Intelligence Framework [1, 4, 7, 15]
- Policy Intelligence Dashboard-
- (Knowledge Management) [18, 33, 78, 79]
- Policy Intelligence [46-49, 80, 81]
- AI Governance (Ethical) [1, 5-7, 15]

-
- AI Governance Framework
- Policy Intelligence- (Pilot Analytical Capabilities)

: ()

-
-

- AI Governance [35, 36, 76]
- [24, 35, 77]
-
- (Policy Briefs)
- (Knowledge-Sharing)

-
- AI Governance
-

: ()

-

-
- Bangla Policy Foundation Model (BPFM) [41-50]
- Foundation Model Retrieval-Augmented Generation (RAG)- Policy Analytics [38-42]
- (Public Sentiment) Policy Intelligence [80, 81]
- Startup Ecosystem- [16, 27, 35, 36]
- [19, 24, 27]

- (Research Outputs)
-
-

: ()

-
-

- (Comprehensive Program Evaluation)
- (Five-Year Impact Assessment)
- Governance Operational Framework [1, 9, 15]
-
- [19, 24, 27]

- Policy Intelligence Framework
-
-

(Success Indicators)

:

-
- Policy Study
-
-
-

- (Knowledge)
-

[9, 13, 20, 21, 26, 77]

(Risk and Mitigation Considerations)

:

-
- [73, 77]
- [18, 31]
- - [33, 78, 79]
- [17, 20, 21]

:

- (Diversified Funding Strategy)
- [35, 36, 76]
- [19, 24, 27]
- (Incremental Implementation)
- (Continuous Evaluation and Adaptation) [1, 9, 15]

- , BAPIL-
[29, 30]

:

- Policy Intelligence Platform
- AI [46-49]
- AI Governance- [1, 4, 27]
-
- AI Governance Public Sector Innovation (Centre of Excellence)

BAPIL- (Structured)
Intelligence , BAPIL AI Governance, Policy
[1, 4, 5, 17, 21, 28-30]
(Evidence-Based) [1, 3, 9,
22, 29, 30] AI Governance,
,

: (-)

Bangladesh AI Policy & Innovation Lab (BAPIL)-
, [3, 21, 23, 29, 30]
AI Governance, Policy
Intelligence,
; [1, 17, 21, 24, 29, 30] (Resilient)

BAPIL, Artificial Intelligence (AI),
Data Analytics Evidence-Based Policy Intelligence,
[21, 23, 29, 30, 80]

(Strategic Pillars) :

- (Knowledge-Based Governance)
- AI (Responsible AI Ecosystem)
- (National Research Capacity)
- (Human Capital Development)
- (International Collaboration)

(Knowledge-Based Governance)

, (Digital Intelligence)
[11, 17, 18, 20, 21, 57, 58]

:

- (Evidence-Informed Policymaking)
- Policy Intelligence Platform
-
-
-

(Human Oversight)
[1, 5, 11, 17, 21, 68, 69]

AI (Responsible AI Ecosystem)

AI, AI Governance Framework- [1, 4, 5, 6,
7, 15]
:

- AI
- Governance
- AI
-
-

9, 14, 15, 50] , [1, 4, 5, 6, 7, 8,

AI

AI [37, 39, 41, 46, 48, 49]

:

- AI
- Policy Intelligence
- AI-
-
-

[3, 24, 46, 47, 48,

49, 76, 81]

[16, 24, 27,

70, 77]

:

- AI
-
-
- AI Governance Centre of Excellence
- (Technology Transfer)

[16, 24, 35, 36, 70, 77, 80]

AI [35, 36, 73, 76, 77]

:

- AI
-
-
-
- AI Governance

[35, 36, 63, 64, 73, 76]

[10, 17, 20, 21, 32, 57, 61]

:

-
- Policy Analytics Platform
-
-
-

;

[10, 11, 17, 20, 21, 32, 52, 57, 61, 68, 69, 82]

AI Governance , [1, 15, 19, 27]

:

- AI Governance
- Digital Government
- Responsible Innovation
- Cybersecurity
- Emerging Technologies

, [15, 19, 27, 54, 55, 60, 65, 70, 75]

BAPIL-

:

-
-
-
-
-
-

,

[3, 5, 22, 23, 24, 74]

:

- AI
- AI
-
- AI
-
- Responsible AI
- AI Governance

[20, 21, 26, 71, 73, 77, 80]

Artificial Intelligence, Digital Governance, Policy Intelligence [21, 29, 30, 80], BAPIL AI [1, 3, 4, 5, 17, 23, 24], [19, 27, 35, 36, 70, 75]

: (Responsible AI)

(Artificial Intelligence AI) (Decision Support) [1, 10, 17, 21, 24] AI (Operational) [2, 5, 6, 7, 15, 41, 45] AI (Responsible AI) Governance Framework, (Human Oversight), (Transparency), (Accountability) [1, 4, 5, 7, 15] AI- [6, 10, 21] BAPIL Framework- AI- Responsible AI-

AI (Responsible AI) ?

AI (Ethical), (Transparent), (Accountable), (Safe) (Human Values) [1, 4, 5, 7, 15] Responsible AI- ; [1, 5, 6, 74] (Potential Harm)

BAPIL Framework- Responsible AI (Foundational Principle)
[1, 4, 5, 7, 15]

(Ethical Principles)

BAPIL Framework AI Governance Framework-
[1, 4, 5, 6, 7, 15, 51]

(Human-Centered Design)

AI (Meaningful Human Oversight)
[1, 5, 6, 15, 68]

:

- (Human Accountability)
- (Expert Review)
-
- (Excessive Automation)

(Transparency)

AI- [1, 4, 5, 7, 50, 51]

:

- (Clear Documentation)
- (Explainable Methodologies)
- (Disclosure of Limitations)
- (Assumptions)

(Fairness)

AI [1, 5, 6, 7, 45, 74]

:

- (Bias)
- (Inclusive)
-
- (Accessibility)

(Accountability)

AI- [1, 6, 15, 68, 69]

:

- Governance-

- (Audit)
- (Review)
-

(Privacy Protection)

, [5, 7, 14, 18, 34]

:

- (Data Minimization)
- (Secure Storage)
- (Access Controls)
- (Regulatory Compliance)

AI-

AI, ,
[2, 6, 7, 41, 45]

(Bias) **(Discrimination)**

AI (Design) [5, 41, 45]

:

-
-
-

[5, 7, 45]

(Lack of Transparency)

AI [42, 43, 50]

:

-
-
-

Explainable AI (XAI) AI [1, 5, 7, 50]

(Privacy and Data Protection Risks)

Policy Intelligence System [18, 33, 78, 79]

:

-

- (Data Breach)
-

Data Governance Information Security [7, 14, 18, 34]

(Security Risks)

AI :

- (Cyberattacks)
- Adversarial Manipulation
-
- (Data Corruption)

(Lifecycle) Cybersecurity [2, 7, 14, 71, 72]

(Overreliance on Automation)

, AI - [1, 5, 15, 68]

:

-
- AI-
-

- [1, 5, 15]

(Misinformation and Information Integrity)

AI (Synthetic) , , [39, 42, 43]

:

- (Deepfake)
- (Automated Misinformation)
-
-

, [2, 5, 22, 45, 74]

(Governance and Oversight Mechanisms)

AI Governance Structure [1, 4, 7, 15]

:

- (Ethical Review)
- (Risk Assessment)

- (Independent Oversight)
- (Transparency Reporting)
- (Continuous Monitoring)
- (Stakeholder Consultation)

Governance AI [1, 5, 7, 15, 51, 53]

Policy Intelligence- Responsible AI

Policy Intelligence System- , [10, 17, 21]

:

-
-
- AI-
-
-

AI- ; [1, 5, 7, 10, 61]

(International Best Practices)

BAPIL Framework , :

- OECD AI Principles [1, 4]
- UNESCO Recommendation on the Ethics of AI [5]
- NIST AI Risk Management Framework [7]
- European Union AI Act [6]
- Singapore Model AI Governance Framework [51, 53]

Framework , [1, 4, 5, 6, 7, 15, 51]

(Building Public Trust)

AI (Public Trust) [1, 4, 5]

:

-
-
- Governance
-
-
-

AI , [1, 4, 5, 15, 74]

BAPIL-

BAPIL- Responsible AI (Cross-Cutting) ,
, Governance, [1, 4, 5, 24]

BAPIL- :

- AI Ethics AI Governance
- Responsible AI
-
-
-

[1, 5, 15, 24, 75]

Artificial Intelligence , [2, 10, 17, 21] AI
, ; ,
, Governance [1, 4, 5, 6, 7, 15]

, , , , BAPIL Framework
Responsible AI Ecosystem ,
[1, 5, 7, 14, 23, 24]

White Paper- (Key Findings)
AI Governance, Policy Intelligence Public Sector Innovation-
BAPIL-

:

Bangladesh AI Policy & Innovation Lab (BAPIL) -
(Policy Intelligence) , (Artificial In-
telligence - AI) , (Evidence-based)
[1,3,10,17,21]

,
, AI ,
[1,5,21,29,30]
;

: **AI**
 (AI) , - ,
 AI [1,4,5,6,7]

- :
- (Transparency) (Explainability) ;
 - (Oversight) ;
 - (Risk-based) ;
 - AI (Ethical AI)- ;
 - (Human Oversight)- ;
 - (Stakeholder)

OECD AI Principles, UNESCO Recommendation on the Ethics of AI, NIST AI
 Risk Management Framework European Union AI Act-
 [1,4,5,6,7,15,51] , ,
 AI , [1,5]

: **AI**
 [37,39,44]

- :
- Natural Language Processing (NLP)- ;
 - ;
 - Large Language Model (LLM) Foundation Model ;
 - (Speech) AI ;
 - (Open) ;
 - AI
- AI (Digital Inclusion),
 [24,46,47,48,49,81]

: **(Open Data)** -
 , [18,21]

- :
- (Open Government Data) ;
 - ;
 - (Interoperability) ;
 - (Research Repository) ;
 - -
- , , [18,21,33,78,79,82]

: **(National Knowledge Repository)**
 [10,11,17,32]
 :
 • - (Policy Repository);
 • ;
 • Knowledge Graph ;
 • Semantic Search ;
 • (Institutional Memory)
 [10,11,17,21,32,58]
 : **AI**
 AI
 [16,35,36,73,77]
 :
 • (Research Grants);
 • (Graduate Scholarships);
 • ;
 • ;
 • AI ;
 •
 [16,27,35,36,70,77,80]
 : **AI (Responsible AI Adoption)**
 (AI) , (Oversight) (Risk Manage-
 ment) [1,5,6,7]
 :
 • AI (Ethical AI) ;
 • (Transparency) ;
 • (Bias) (Fairness) ;
 • (Privacy) ;
 • (Cybersecurity) ;
 • (Monitoring)
 AI [1,5,6,7,14,15,50,74]
 AI (Human
 Oversight) AI- , , ,

: - - AI
 , [3,19,27]
 :
 • ;
 • ;
 • ;
 • - (Public-Private Partnership);
 • - ;
 • (Policy Dialogue)
 [19,27,35,36,70,80]
 : AI (AI Literacy)
 AI- (AI Readiness) , , ,
 [26,73,76,80]
 :
 • AI ;
 • (Digital Literacy) ;
 • AI ;
 • AI ;
 • ;
 • -
 AI , AI- ,
 [35,36,64,73,76,80]
 : (Digital Resilience)
 ,
 [14,34,71,72]
 :
 • ;
 • (Critical Infrastructure) ;
 • (Information Integrity) ;
 • ;
 • (Incident) ;
 • (Cyber Hygiene)
 [14,34,71,72]

: **AI**
 AI
 [16,19,27,35,36]
 :
 • ;
 • AI ;
 • AI ;
 • - ;
 • ;
 •
 [19,27,54,55,60,65,70] , AI AI
 AI ,
 [1,17,21,24,29,30]
 , ,
 [29,30,80]
 , AI-
 , [1,5,23,24,73,80]
 :
 (Artificial Intelligence - AI)
 [1,3,10,17,21,22,23]
 AI-
 [10,17,20,21]
Digital Bangladesh **Smart Bangladesh-**
 [28,29,30,32,82,83]
 [17,21,24,29,30]
 ,
 [2,5,7,14,71,72,73,80]
Bangladesh AI Policy & Innovation Lab (BAPIL)-
 , Policy Intelligence,
 [1,10,17,21,25] BAPIL

; (Multidisciplinary) ,
 , AI [3,24,25,27]
 :
 • AI Policy Intelligence ;
 • (Public Sentiment) ;
 • AI ;
 • ;
 • AI , ;
 • ;
 • AI (Knowledge Systems)
 , , , , (UAE)
 , AI
 [51,52,54,57,58,60,61,62] (Conceptual Architecture),
 [29,30]
AI (Responsible AI) ;
 ;
 Oversight) [1,4,5,6,7,15] AI- (Human
 [4,5,15]
 Policy Intelligence , AI
 [35,36,46,48,49,80] ,
 AI
 [16,27,35,36,73,77]
 , **BAPIL-** ,
 [1,3,17,24,25,27] ,
 [5,22,23,24]
 , AI ,
 [22,23,29,30]
 ;
 , [1,5,22,24] ,
 AI ,
 [4,5,23,24,29,30]

A:

Bangladesh AI Policy & Innovation Lab (BAPIL)-
[10,17,21,24]
, Policy Intelligence (Knowledge Man-
agement) ,
[10,17,18,21]
(Conceptual)
(Operational)
:
**(Bangladesh Policy Search En-
gine)**

, , (Semantic Search)
(Knowledge Discovery) [18,33,38,57,59]

- (Natural Language) [38,42]
- (Semantic Document Retrieval) [38]
- AI- (Document Summarization) [39,42]
- (Cross-document) (Citation)
- (Knowledge Graph) [18]
- (Trend)

- - [33,83]
- (Knowledge Management) [57,58]
-
- [17,21]

: **(Budget Intelligence Dashboard)**

, , (Analytical)
[17,18,31]

-
-
-

-
- [17,20]
-

-
- [17,21]
-
- (Fiscal) [18]

: (Citizen Feedback Analyzer)

, [21,24,25], AI-

- (Sentiment Analysis) [81]
- (Topic Modeling)
-
- [81]
-
-

-
- [18]
- [21,25]
-

: (Bangla Policy Foundation Model
- BPFM)

Model), [37,39,41,46,48,49] (Foundation

- - (Policy Question Answering) [39,42]
- [42]
- (Semantic Search and Retrieval) [38]
- Retrieval-Augmented Generation (RAG) [38]

-
- [46,48,49]

- AI [46,48,49]
- -
-
-

: AI (National AI Readiness Observatory)

, , AI [26,77,80]

- AI (AI Readiness) [26,80]
- [73,77]
-
-
- (Benchmarking) [26,77]
- AI

- AI
- AI [26,80]
-
-

, AI [10,17,21], BAPIL (Policy Intelligence), [1,7,17],

AI ,

B:
Concept)

(International Advisory Board

AI , , (Policy Intelligence)
, (Best Practices) BAPIL
, - (Policy Communities)
[1,3,5,10,17,27]

 , ,
 ,

OECD	AI , , [1,4,10,11,13]
UNESCO	AI , [5,67,76]
NIST	AI , [7,14]
World Bank	GovTech, [17,18,19,20]
UNDP	, [3,24,25]

- :
- AI [1,4,5]

• AI (Responsible AI) [5,7]

• GovTech [17,20,21]

• [10,17]

• [25,76]

• (Policy Intelligence)

• AI [46,48,49]

• [17,27]

•

•

•

BAPIL :

- OECD AI Principles [1,4]

• UNESCO Recommendation on the Ethics of Artificial Intelligence [5]

• NIST AI Risk Management Framework (AI RMF) [7]

• European Union AI Act [6]

- | | | |
|---------------------|----------------|---------------|
| (Trustworthy), | (Transparent), | (Accountable) |
| (Human-Centered) AI | | [1,4,5,7] |

(Disclaimer)

C: (Funding Model)

(Funding Philosophy)

$$\vdots$$

- AI, (Research Integrity)
[1,5,7,15,24]

98

•

AI , (Responsible Innovation)
[1,17,24,37-50,70,77].

(Human Capital Development)

•

•

•

•

AI (AI Readiness)
[35,36,63,64,73,76,77,80].

(Technical Infrastructure)

•

•

•

•

(Knowledge Management)

AI
[18,20,21,33,78,79].

(Collaboration and Outreach)

•

•

•

•

(Multi-stakeholder Engagement)
AI [3,19,24,25,27,75].

(Sustainability Strategy)

:

•

•

•

•

•

•

(Policy Intelligence) , AI ,
[16,17,19,24,27,35,36].

(Potential Development Stages)

: **(Independent Research Initiative)**

-
-
-
-

[17,24,27]. , Proof-of-Concept

: **(Pilot Research Programs)**

-
- (Policy Intelligence)
- (Knowledge Repository)
-

[10,17,21,24,38]. , AI-

: **(Institutional Partnerships)**

-
-
-
-

[17,19,24,27,35,36]. , (Knowledge Transfer)

: **(Mature Research Ecosystem)**

-
-
-
-

[17,24,27,29,30]. ,

(Disclaimer)

, , BAPIL-
 , ,

(Acknowledgements)

OECD, UNESCO, UNDP, United Nations,
World Bank, World Economic Forum
[1-27,71-77]

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